

Perpetual vs Spot: Is Bitcoin Perpetual Funding a Harvestable Carry?

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Abstract

A long position on a perpetual futures contract continuously pays (or receives) *funding*. Viewed as a cost on a leveraged directional bet, funding merely erodes returns; viewed as a cash flow on a market-neutral book, it is a carry. Using daily Binance data for BTC, ETH and SOL over 2021-01-01 to 2024-12-31, this study compares three books — a spot long, a 4× leveraged perpetual long, and a delta-neutral cash-and-carry (long spot, short perpetual) — on a daily mark-to-market basis with a maintenance-margin liquidation guard. Over a walk-forward across 16 non-overlapping quarters, the 4× perpetual long has a *negative* median quarterly return (-3.1%) and is liquidated in 3 quarters, whereas the delta-neutral carry is profitable in every quarter (median +2.2%, median Sharpe 9.09, worst drawdown -0.5%) by harvesting a funding rate that averages 13.6% annualized. The central finding is that, on a risk-adjusted and out-of-sample basis, the edge is the funding carry, not the leverage.

1 Introduction

Perpetual futures (“perps”) are the dominant instrument in crypto derivatives. Unlike dated futures they never expire; instead, a periodic *funding* payment exchanged between longs and shorts tethers the perpetual price to spot. When the perp trades at a premium, longs pay shorts, and vice versa. A natural question for a directional trader is how much this funding erodes a leveraged long. A more interesting question for a systematic desk is whether the funding stream is itself a tradable source of return.

This report answers both. We frame three books and evaluate them not on a single hand-picked window but across multiple market regimes, with risk-adjusted metrics and an explicit liquidation model. We deliberately avoid two common pitfalls: (i) arguing “market integration” from the correlation of price *levels*, which is spurious for trending series [1, 2]; and (ii) comparing raw returns across strategies that run at different leverage.

2 Data

All series are daily and sourced from Binance’s public, geo-neutral historical archive (data.binance.vision): spot and USDT-margined perpetual klines, and the 8-hour funding rate joined to the mark price at each settlement. The snapshot covers 3 symbols (BTC, ETH, SOL) over 2021-01-01–2024-12-31 (1,461 daily bars per symbol, three funding settlements per day), and is committed to the repository so every result reproduces offline. BTC is the primary symbol; ETH and SOL are used to test generalization.

3 Methodology

Market integration. Let $b_t = P_t^{\text{perp}} - P_t^{\text{spot}}$ be the basis. Integration is assessed by (i) the correlation of daily *returns*; (ii) the basis in basis points, its mean-reversion half-life from an

AR(1) fit, and an Augmented Dickey–Fuller (ADF) stationarity test; and (iii) an Engle–Granger cointegration test on the two price levels [3].

Books. With investment (margin) M and leverage L , the perpetual position size is $q = ML/P_0^{\text{perp}}$ asset units. Each book is marked to market daily:

- **Spot** $1\times$: buy M/P_0^{spot} units, hold.
- **Perp** $L\times$: equity = $M + q(P_t^{\text{perp}} - P_0^{\text{perp}}) - \text{funding}_t - \text{fees}$, with $L = 4$.
- **Carry (Δ -neutral)**: long M/P_0^{spot} units of spot, short the same notional of perp. Net price exposure is the basis only; the short perp leg *receives* funding.

Funding accrual. Funding is summed strictly inside the holding window $(t_{\text{open}}, t_{\text{close}}]$, charged on the live notional:

$$\text{funding} = \sum_i r_i \cdot P_i^{\text{mark}} \cdot q,$$

where r_i is the 8-hour rate. A positive rate is a cost for a long and income for the short carry leg; the sign is handled by arithmetic, so no regime needs special casing. Reported as an annualized rate, funding is quoted on *notional*, not on the posted margin.

Risk and liquidation. Metrics are computed on the daily equity path: annualized Sharpe and Sortino, maximum drawdown, Calmar, historical VaR/CVaR and hit rate. A maintenance-margin guard liquidates the leveraged book (capping its loss at the margin) whenever equity falls below $m \cdot$ notional, so the path risk that endpoint-only P&L hides becomes visible.

Out-of-sample. Books are run over 16 non-overlapping calendar quarters spanning bull, bear and sideways regimes; results are reported as distributions, with a stationary block bootstrap for confidence intervals.

4 Market structure

Correlating the two price *levels* gives 0.999999, but this is mechanical: both series share a trend, so the number is near one regardless of how tightly the markets are linked. The economically meaningful quantity, the correlation of daily *returns*, is 0.999964. Integration is better shown by the basis itself: it averages -1.03 bp (s.d. 5.41 bp) with a mean-reversion half-life of 4.8 days, ADF rejects a unit root ($p = 3.2 \times 10^{-5}$), and the price levels are cointegrated ($p = 0.013$). Daily returns are emphatically *not* Gaussian: excess kurtosis is 3.41 and Jarque–Bera gives $p = 1.8 \times 10^{-155}$ — fat tails that make naive volatility and VaR understate downside, and that justify the liquidation model for leverage. Annualized volatility is 62.1%.

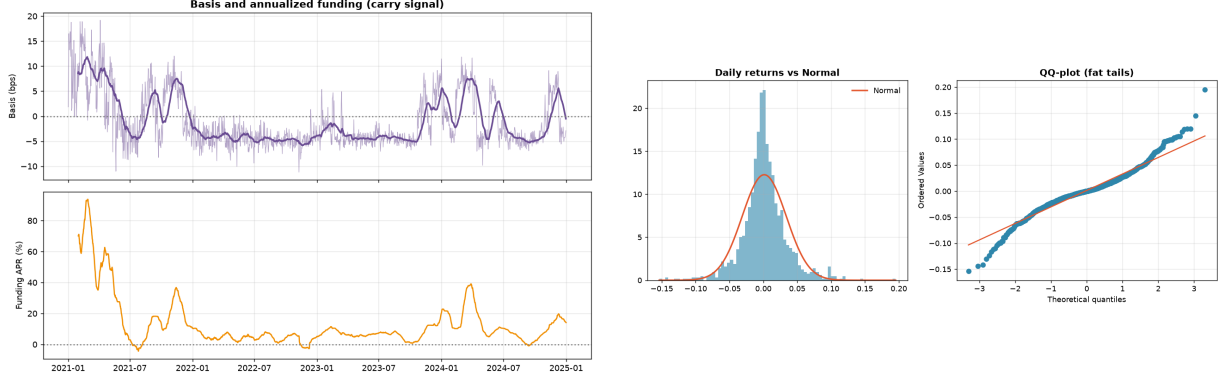


Figure 1: Left: the perp-spot basis (top) and rolling annualized funding (bottom), a positioning/carry signal. Right: the daily-return distribution and QQ-plot, showing heavy tails relative to the normal.

5 Strategy results

5.1 A single-window case study

Over the 2024-Q1 window (2024-01-01 to 2024-03-20, $M = \$1,000$, $L = 4$):

Strategy	Return	Sharpe	Max DD	Funding
Spot $1\times$	+60.2%	3.78	-15.8%	\$0
Perp $4\times$	+215.2%	3.85	-50.5%	\$261
Carry (Δ -neutral)	+6.2%	15.15	-0.3%	\$65

Table 1: Case-study window. Leverage scales return and volatility together, so the $4\times$ Sharpe ($\approx$ spot) buys no risk-adjusted edge while drawdown scales $\sim 3\times$. Funding would need to reach 169.5% APR to erase the leverage edge in this window — yet out of sample (below) leverage still loses.

5.2 The carry across the full history

Held continuously over 2021-01-01–2024-12-31, the delta-neutral carry returns +94.2% (Sharpe 8.84, max drawdown -0.3%), accumulating \$943 of funding income on \$1,000 of capital with near-zero price exposure.

5.3 Out-of-sample, across regimes

Across all 16 non-overlapping quarters:

Strategy	Median	Worst	Med. Sharpe	Worst DD	Liquidations
Carry (Δ -neutral)	+2.2%	+0.4%	9.09	-0.5%	0 / 16
Perp $4\times$	-3.1%	-100.0%	1.08	-100.0%	3 / 16
Spot $1\times$	+3.0%	-56.4%	0.48	-59.4%	0 / 16

Table 2: Walk-forward distribution of quarterly outcomes. The $4\times$ perp has a negative median and is liquidated in 3 of 16 quarters; the carry is tight, positive and never liquidated.

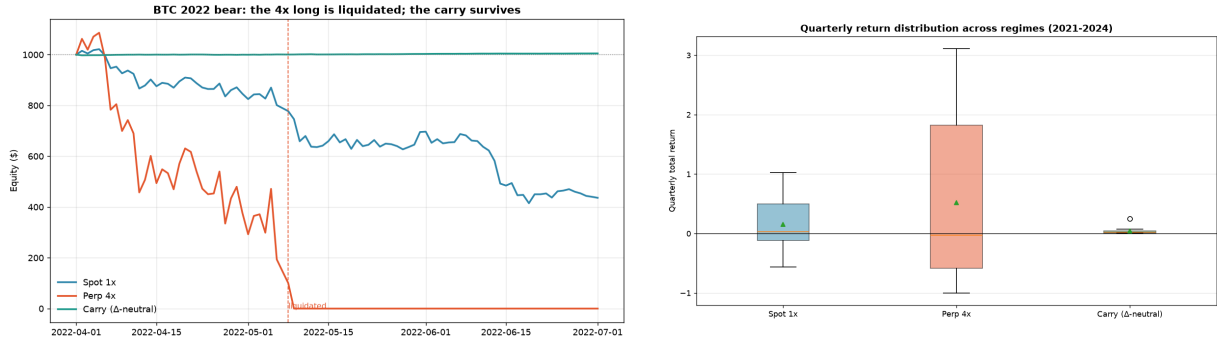


Figure 2: Left: the 2022 bear quarter — the $4\times$ long is liquidated while the carry is unscathed. Right: the distribution of quarterly returns by strategy; the carry’s box is a tight sliver above zero.

5.4 Generalization to ETH and SOL

The pattern is not BTC-specific. Median quarterly returns and $4\times$ liquidation counts by symbol:

Symbol	Spot $1\times$	Perp $4\times$	Carry	Perp liq.
BTC	+3.0%	-3.1%	+2.2%	3
ETH	+20.3%	+46.5%	+2.5%	3
SOL	+16.6%	+31.2%	+1.4%	6

Table 3: For high-momentum alts the leveraged long can have a positive median, but it is still liquidated several times; the carry stays positive and is never liquidated.

6 Discussion

Three points follow. First, “leverage beats spot” is true by construction in an up-window — a $4\times$ long’s gross return is mechanically four times the spot return — and reverses in any down-window, so it is not a finding. Second, on a risk-adjusted basis leverage is roughly Sharpe-neutral (it scales numerator and denominator together) while it multiplies drawdown and adds liquidation risk; unconditionally, the $4\times$ book is a coin flip that blows up about one quarter in five. Third, the funding stream the directional trader pays is, for a market-neutral book, a double-digit-APR carry harvested at a fraction of the directional drawdown and with near-zero beta. The positive, persistent funding is itself a signal: it reflects crowded longs paying to be long.

7 Limitations

Funding APR figures assume the 8-hour schedule (exact for BTC; some ETH/SOL periods used a 4-hour schedule and are approximate). The liquidation guard is a simplified flat maintenance-margin check, not a tick-level engine. Round-trip taker fees are charged on notional, but slippage, partial fills, borrow constraints on the spot leg, and the funding impact of the position itself are not modelled; the carry assumes the short-perp leg is always fundable. Results are historical and regime-dependent — funding compresses, and can invert, in bear markets.

8 Conclusion

Comparing a spot long, a leveraged perpetual long and a delta-neutral funding carry across multiple regimes and three assets, with daily mark-to-market and an explicit liquidation model,

the leveraged long offers no risk-adjusted edge and is periodically wiped out, while the delta-neutral carry turns the same funding stream into a stable, double-digit-APR return. The edge is the carry, not the leverage.

References

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